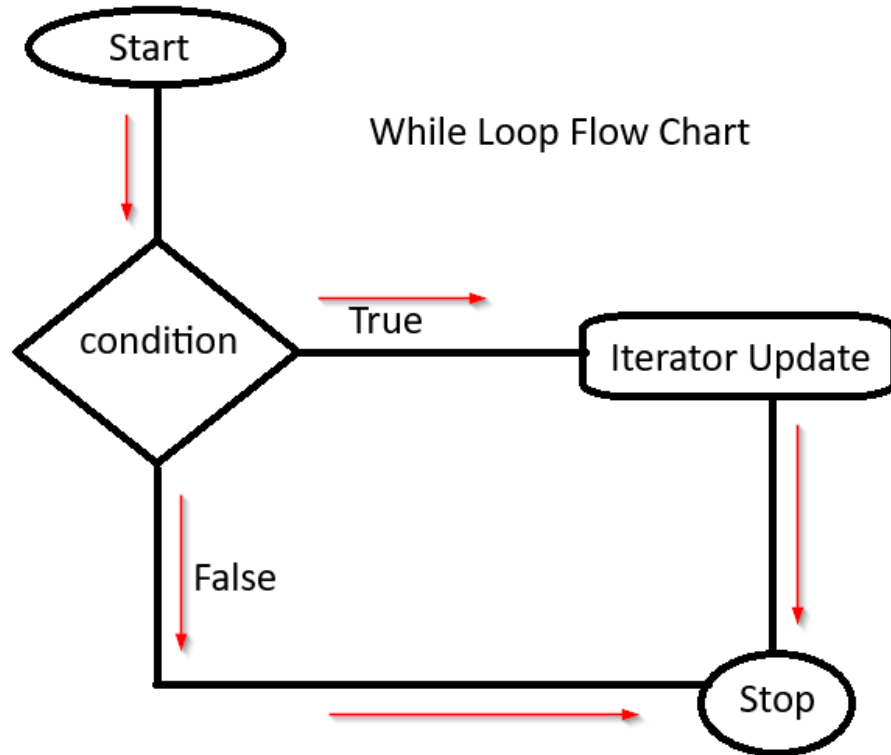


13_While Loop



```
# To perform a task multiple times for a given set of inputs or entries then we will go for loops (While loop, For loop)
# In While loop, the body will be continuously executed until the condition is true
# Iteration over each input/entry should be stopped at one point either by updating the iterator in the loop or condition failure at one point

counter = 1
while counter < 10:
    counter += 1
    print("The Number updated to ",counter)
```

[2] ✓ 0.0s

```
... The Number updated to 2
The Number updated to 3
The Number updated to 4
The Number updated to 5
The Number updated to 6
The Number updated to 7
The Number updated to 8
The Number updated to 9
The Number updated to 10
```

Similar to Nested If statements, we do have Nested while loops as well



```
# Nested while loop example
teacher_cnt = 1
principal_cnt = 1

while teacher_cnt <= 3:
    print("Teacher's Turn ",teacher_cnt)
    while principal_cnt <= 3:
        print("--> Principal's Turn ",principal_cnt)
        principal_cnt += 1
    teacher_cnt += 1
```

[4] ✓ 0.0s

```
... Teacher's Turn 1
--> Principal's Turn 1
--> Principal's Turn 2
--> Principal's Turn 3
Teacher's Turn 2
Teacher's Turn 3
```



```
num_list = [2,3,5,7,10]
sum_sqaure_nums= 0
index=0

while index < len(num_list):
    sum_sqaure_nums += num_list[index]**2
    index += 1

print("Sum of squares of all given numbers is ", sum_sqaure_nums)
```

[7] ✓ 0.0s

```
... Sum of squares of all given numbers is 187
```

Just like If Else condition, we do have While Loop with Else

```
▷ # While loop Else

teacher_cnt = 1
principal_cnt = 1

while teacher_cnt <= 3:
    print("Teacher's Turn ",teacher_cnt)
    while principal_cnt <= 3:
        print("--> Principal's Turn ",principal_cnt)
        principal_cnt += 1
    else:
        print("--> Principal's Turn already completed")
    teacher_cnt += 1
```

[8] ✓ 0.0s

```
... Teacher's Turn 1
--> Principal's Turn 1
--> Principal's Turn 2
--> Principal's Turn 3
--> Principal's Turn already completed
Teacher's Turn 2
--> Principal's Turn already completed
Teacher's Turn 3
--> Principal's Turn already completed
```

Failure to update the iterator during he iteration makes the loop to run infinite time
Here variable used inside the loop to check the condition is Iterator and the loop execution over the input variables or inputs is nothing but iteration